UNIT :- 3

## (A) Networked Business Systems

It views organizations not as isolated units, but as interconnected networks that exchange materials, information, money, and services.

These networks :-
- → Improve efficiency.
- → Flexibility.
- → Competitiveness.

Network
↓
Interconnections
↓
Vertices → Entities / Nodes
↓
Edge → Links
↓
Data → Flow

## # Multiple Type Of Networks

(i) Supply Network.
(ii) Information Network.
(iii) Project / Progress Network.

## (i) Supply Network : (Amazon)

A supply network (or extended supply chain) links suppliers, manufacturers, distributors, retailers, and customers.

Key Elements :

- Upstream Activities.
- Internal operations.
- Downstream Activities.

characteristics.

- Multi-tier suppliers.
- Global sourcing.
- Risk sharing.
- Coordination of logistics and Inventory.

## (ii) Information Network : (Walmart)

An Information network connects organization and stake holders through data-sharing systems.

Components.

- ERP Systems.
- Cloud platforms.
- EDI (Electronic Data Interchange)
- IoT and Data analytics.

function.

- Real time tracking
- Demand forecasting.
- Collaborative Planning.
- Transparency and visibility.

## (iii) Progress / Project Network :

A Project Network represents project activities and their dependencies in graphical form.

Types:

- Activity-on-Node (AON)
- Activity-on-Arrow (AOA)

Purpose:-

- Identify sequence of tasks.
- Determine critical path.
- Estimate project duration.
- Allocate resources.

## Project Planning fundamentals:

Project Planning ensures systematic execution of tasks to meet time, cost, and scope objectives.

### Activities:

An activity is a specific task that consumes time and resources.

Each activity has:

- Duration.
- Resource Requirement.
- Start and finish time.

Examples:

- Design phase.
- Procurment.
- Testing.
- Deployment.

## (B) Events (Milestones):

An Event marks the Beginning or completion of activities.

**Characteristics:**

- No Duration.
- Used as control points.
- Represent achivement of Key Stages.

**Example:**

- Project Start.
- Prototype approval
- Final Delivery.

## (C) Dependencies:

Dependencies defines relationships between activities.

**Types:-**

- → Finish - to - Start (FS)
- → Start - to - start. (SS)
- → Finish - to - finish. (FF)
- → Start - to - Finish. (SF)

Task A
— when
Task B
— .

## (*) Time - Cost Trade - Off

In projects, time and cost are inversely related.

- Shorter duration → Higher cost.
- Longer duration → Lower cost.

Type of Costs:

1. Direct Costs → Labour, Materials, equipments.
2. Indirect Costs → Supervision, Overhead, penalities.

Date
Page

Project Managers aim to find the Optimal project duration where total cost is Minimized.

## Project Crashing Concepts :

Project crashing is a technique to shorten project duration by adding resource at extra cost.

When Crashing is used :-

- To meet deadlines.
- Avoid penalties.
- Gain market advantages.

Steps in Crashing :

- Identify the Critical path.
- Calculate crash cost per unit time.
- Crash the activity with lowest crash cost first
- Continue until desired duration is achieved.

## Crash Cost Formula :-

$$\text{Cost Slope} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Normal Time} - \text{Crash Time}}$$

## (*) Network - Based Decision Logic

It focuses on identifying constraints, interdependencies, and coordination requirement within business or project systems.

### (A) Bottlenecks

→ A Bottleneck is a stage in a network where capacity is limited and restricts overall system performance.

Key characteristics:

- Lowest capacity in the process.
- Causes delays and queue buildup.
- Determines overall system throughput.

Example :- In a production network like Toyota, if painting capacity is lower than assembly capacity, painting becomes the bottleneck.

### (B) Coordination in Networks

Coordination ensures smooth flow of:

- Materials.
- Information
- Resources.
- Tasks.

## Type of Coordination :

(i) Sequential Coordination :- Tasks follow logical order.

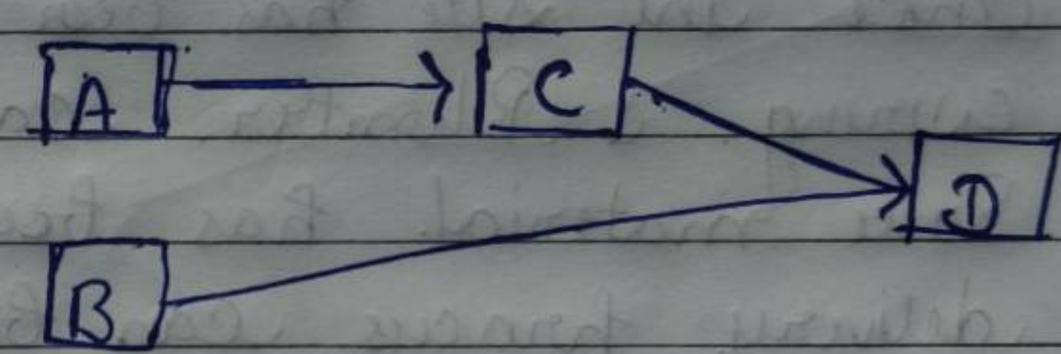
(ii) Reciprocal Coordination :- Activities depends on each other.

(iii) Pooled Coordination :- Independents units contribute to a common goal.

# Draw Networks :-

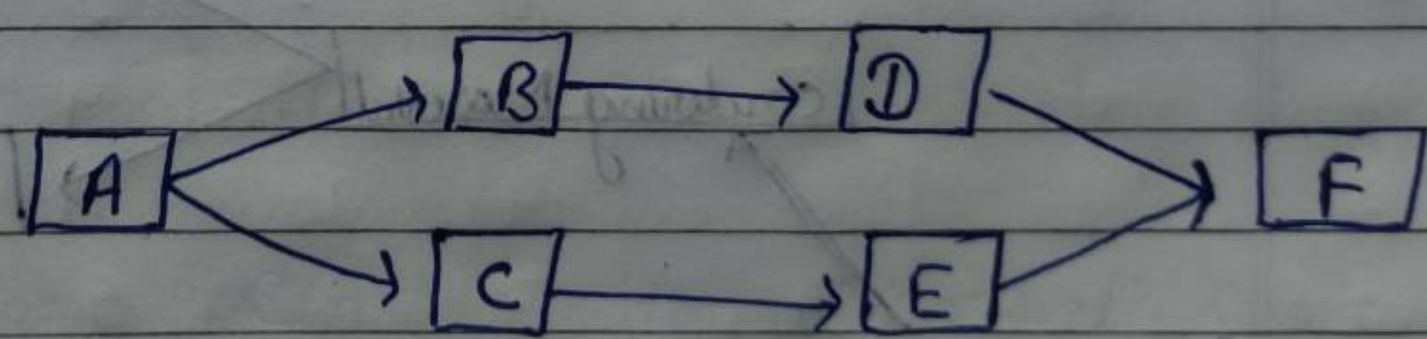
(i)

A
B
C (after A)
D (after B & C)



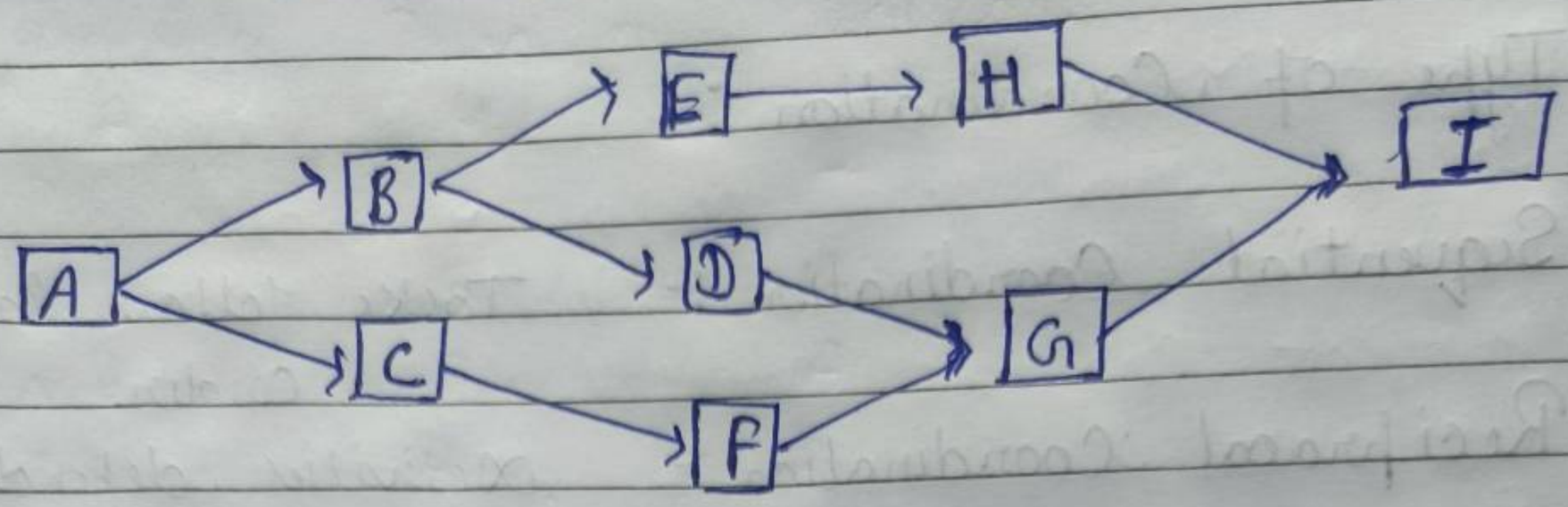

(ii)

| Activity | Predessor |
|---|---|
| A | — |
| B | A |
| C | A |
| D | B |
| E | C |
| F | D&E |

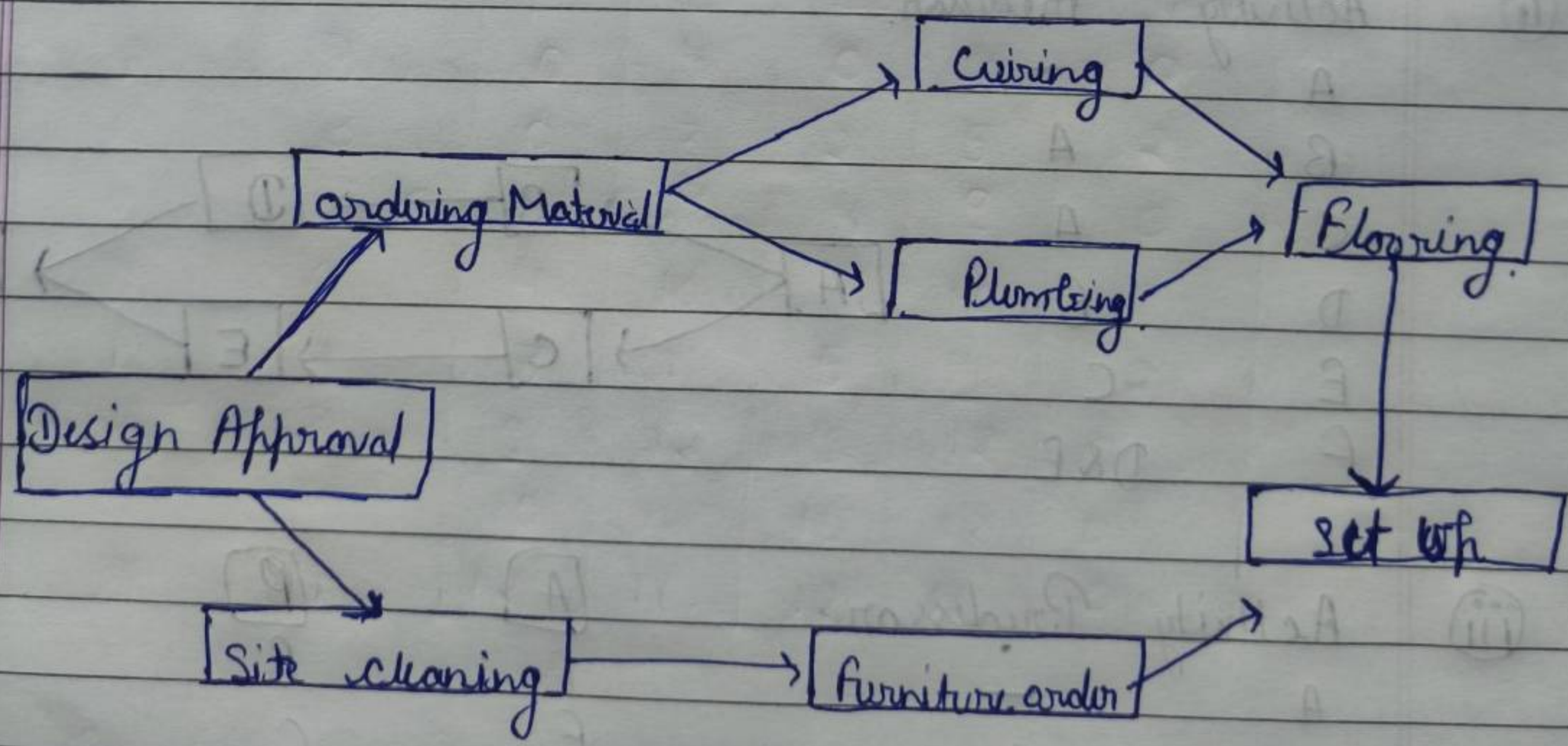



(iii)

| Activity | Predessor | A | B |
|---|---|---|---|
| A | — | E | B |
| B | A | F | C |
| C | A | G | D, F |
| D | B | H | E |
| | | I | G, H |




Q A Project Development for Interior planning of a building is described in following Manner. Firstly, the customer approved design is Mandatary which can be followed by ordering of Materials and site cleaning simultaneously. Once the site has been cleaned the electrical-wiring & Plumber can complete his task. Since material has been ordered. The furniture delivery process can be take place. Before we go with the final set up we need to install flooring (followed by wiring & Plumbing).



## Critical Path Method :

→ It is the longest path through the project network that determines minimum project duration

Steps :-

(i) List all possible paths (Build Networks)

(ii) Compute forward pass
- → ES : Earliest Start
- → EF : Earliest finish.

(iii) Compute Backward Pass.
- → LS : Latest Start
- → LF : Latest finish.

(iv) Slack Interpretation.

↳ Amount of time an activity can be delayed without affecting. Project completion.

$$\text{Slack} = LS - ES = LF - EF$$

Type :-

(a) Total Slack : Delay allowed without affecting project finish.

(b) Free Slack :- Delay allowed without affecting next activity.

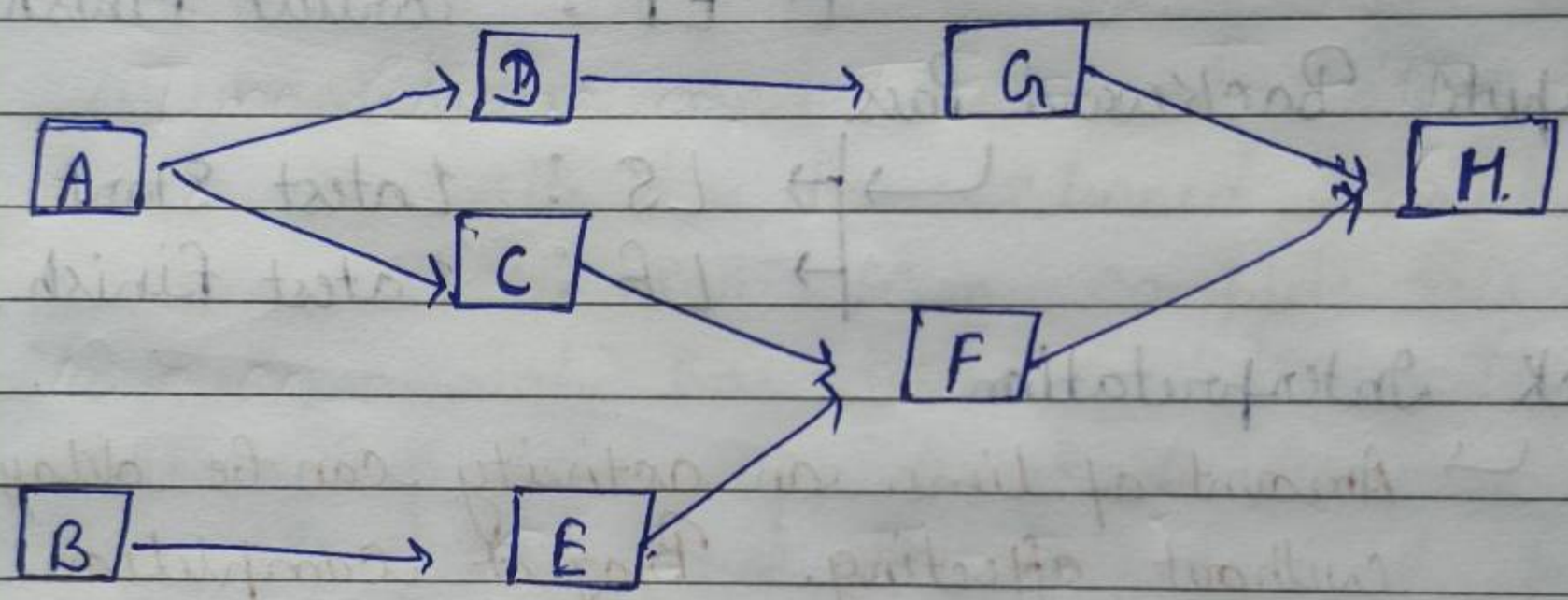
Interpretation :-

- Slack = 0 → Activity is Critical.
- Slack > 0 → Activity has flexibility.
- Higher Slack → Lower scheduling priority.

(v) Identify Critical Path.

Q

| Activity | Duration | Predessar |
|---|---|---|
| A | 4 | — |
| B | 3 | — |
| C | 5 | A |
| D | 6 | A |
| E | 4 | B |
| F | 3 | C, E |
| G | 2 | D |
| H. | 4 | F, G. |




| Predessor | Activity | Duration. | ES | EF | LS | LF | Slack |
|---|---|---|---|---|---|---|---|
| — | A | 4 | 0 | 4 | 0. | 4 | 0 |
| — | B | 3 | 0 | 3 | 2 | 5 | 2 |
| A | C | 5 | 4 | 9 | 4 | 9 | 0 |
| A | D | 6 | 4 | 10 | 4 | 10 | 0 |
| B | E | 4 | 3 | 7 | 5 | 9. | 2 |
| C, E | F | 3 | 9 | 12 | 9 | 12 | 0 |
| D | G | 2 | 10 | 12 | 10 | 12 | 0 |
| F, G | H | 4 | 12 | 16. | 12 | 16 | 0 |

So,

B & E can start after 2 unit of time.

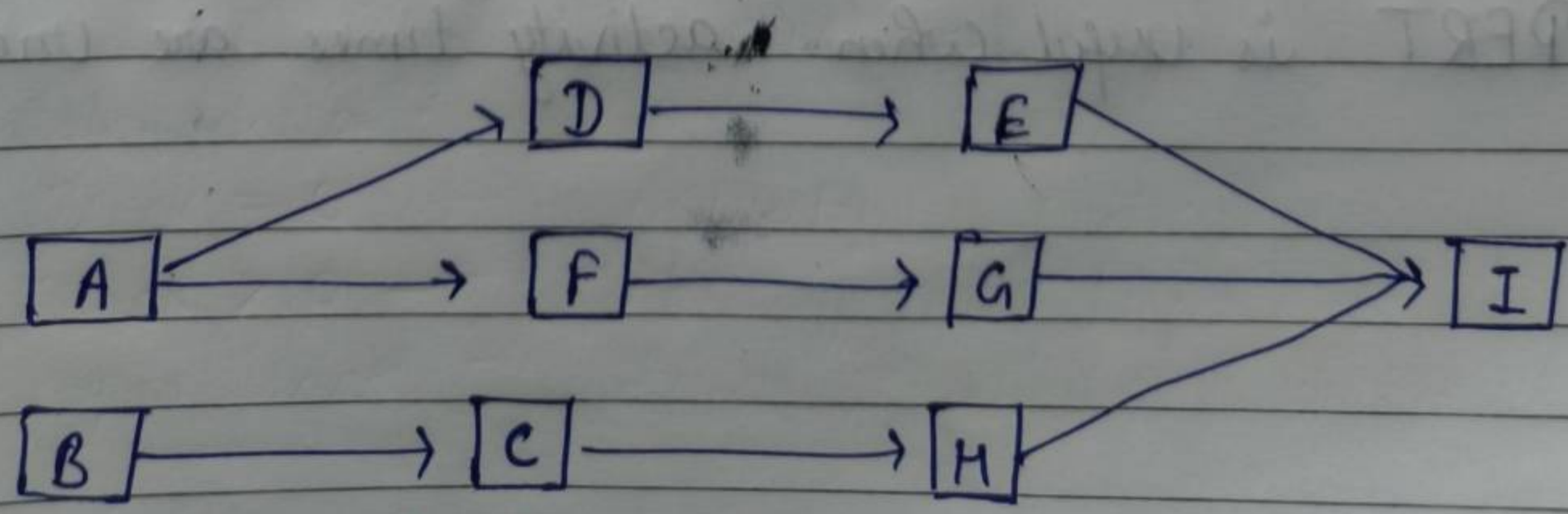
Critical Path. :- A → C → F → H

A → D → G → H.

Q A collage Organise an event in which their they have to conduct some activities:-

(i) To finalize event theame (3 days) (A)
(ii) To prepare sponsarship (4 days) (B)
(iii) Sponsar are connected (After 2nd activity) (5 days) (C)
(iv) Design Poster (4 days) (After theame) (D)
(v) Printing Poster (2 days) (E)
(vi) Simutaneously Booking venue (After theame) (3 Days) (F)
(vii) Arrange Equipment (After Booking) (4 days) (G)
(viii) If sponsar contacted, Availability (2 days) (H)
(ix) Finally Event is executed one we finished (2 days) (I)
(Printing, Arranging equi, Confirmation sponsar)

| Activity | Duration | Precedar | ES | EF | LS | LF | Slack |
|---|---|---|---|---|---|---|---|
| A | 3 | – | 0 | 3 | 1 | 4 | 1 |
| B | 4 | – | 0 | 4 | 0 | 4 | 0 |
| C | 5 | B | 4 | 9 | 4 | 9 | 0 |
| D | 4 | A | 3 | 7 | 5 | 9 | 2 |
| E | 2 | D | 7 | 9 | 9 | 11 | 2 |
| F | 3 | A | 3 | 6 | 4 | 7 | 1 |
| G | 4 | F | 6 | 10 | 7 | 11 | 1 |
| H | 2 | C | 9 | 11 | 9 | 11 | 0 |
| I | 2 | E, G, H | 11 | 13 | 11 | 13 | 0 |




CP :- B → C → H → I

## PERT :

→ Program Evalution and Review Techniques.

→ Network Construction

→ Time Estimation.

· Uses three time to estimates.

(i) Optimistic ($T_o$) (O)

(ii) Most likely ($T_m$) (M)

(iii) Pessimistic ($T_p$) (P)

Time Estimation, $$\boxed{TE = \frac{T_o + 4T_m + T_p}{6}}$$

Variance :-

$$\sigma^2 = \left(\frac{T_p - T_o}{6}\right)^2$$

Standard Deviation :-

$$\sigma = \left(\frac{T_p - T_o}{6}\right)$$

→ PERT is useful when activity times are uncertain.

Q

| Activity | Dependency | O | M | P |
|---|---|---|---|---|
| A | – | 2 | 4 | 6 |
| B | – | 1 | 3 | 5 |
| C | A | 2 | 3 | 8 |
| D | B | 2 | 4 | 6 |
| E | C, D | 3 | 5 | 9 |

find C.P., $\sigma^2$, $\sigma$

| TE | O | M | P | Activi | Depend | Duration | ES | EF | LS | LF | Slack |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 4 | 2 | 4 | 6 | A | – | 4 | 0 | 4 | 0 | 4 | – |
| 3 | 1 | 3 | 5 | B | – | 3 | 0 | 3 | 1 | 4 | 1 |
| 3.66 | 2 | 3 | 8 | C | A | 4 | 4 | 8 | 4 | 8 | – |
| 4 | 2 | 4 | 6 | D | B | 4 | 3 | 7 | 4 | 8 | – |
| 5.33 | 3 | 5 | 9 | E | C, D | 6 | 8 | 14 | 8 | 14 | – |

Duration :- $\boxed{TE}$

$$A = \frac{2+4\times4+6}{6} = 4$$

$$B = \frac{1+4\times3+5}{6} = 3$$

$$C = \frac{2+4\times3+8}{6} = 3.66$$

$$D = \frac{2+4\times4+6}{6} = 4$$

$$E = \frac{3+5\times4+9}{6} = 5.33$$

Variance :- Var A + Var B + Var C + Var D + Var E.

$= 0.44 + 0.44 + 1 + 0.44 + 1$

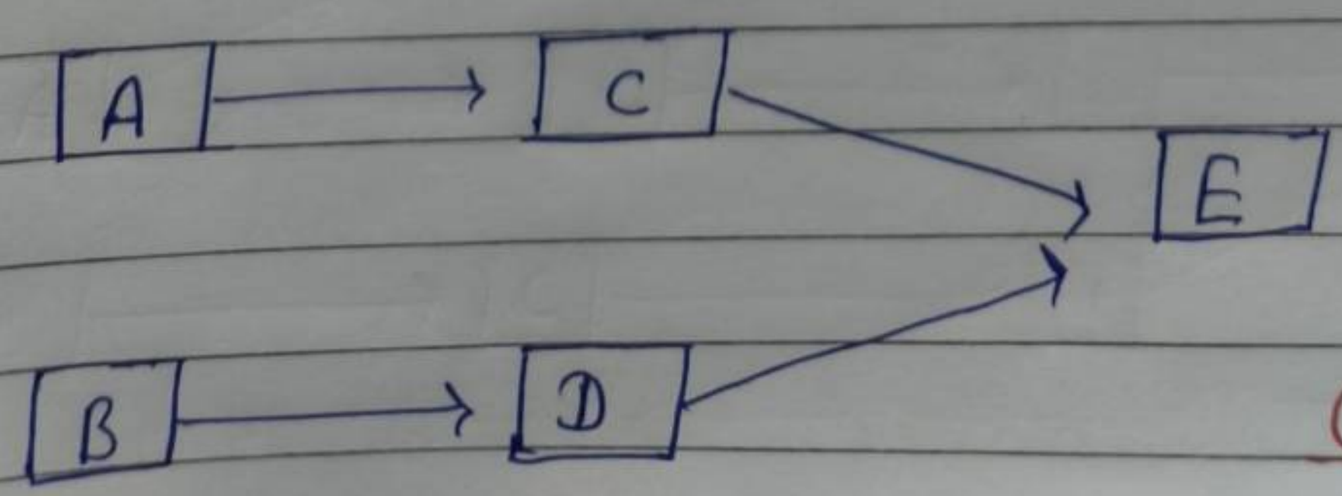
$\boxed{\sigma^2 = 3.32}$

$\boxed{S.D.}$ :- $\sqrt{\sigma^2} = \sigma$

$\sigma = \sqrt{3.32}$

$= 1.82$

Network :-




C.P. :- A → C → E

## Trade-Off & Project crashing

$$\text{Cost Slope} = \frac{CC - NC}{NT - CT}$$

CC = Crash Cost
NC = Normal Cost
NT = Normal Time.
CT = Crash Time.

Ex:-

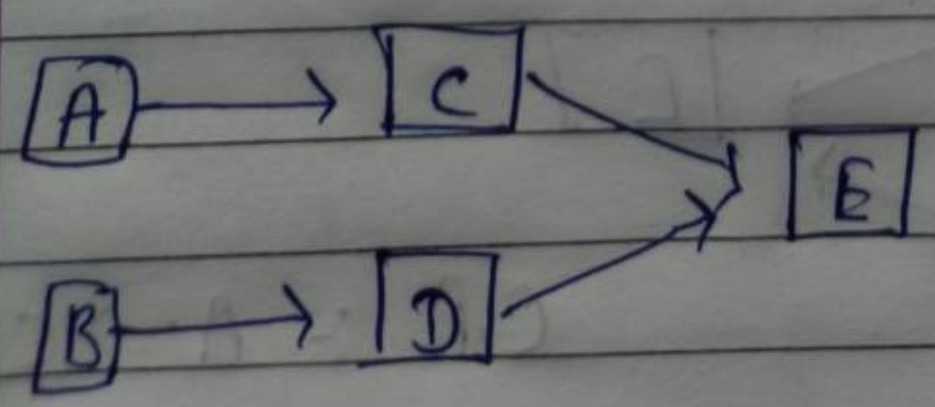
| Activity | Predecessor | NT | CT | NC | CC |
|---|---|---|---|---|---|
| A | – | 6 | 4 | 600 | 1000 |
| B | – | 5 | 3 | 500 | 900 |
| C | A | 4 | 2 | 400 | 700 |
| D | B | 6 | 4 | 600 | 1000 |
| E | C, D | 3 | 2 | 300 | 500 |

first we finding the Critical Path.

→ NT is our duration for activities.

| Acti | Prec | NT | CT | NC | CC | ES | EF | LS | LF | Slack |
|---|---|---|---|---|---|---|---|---|---|---|
| A | – | 6 | 4 | 600 | 1000 | 0 | 6 | 1 | 6 | 1 |
| B | – | 5 | 3 | 500 | 900 | 0 | 5 | 0 | 5 | – |
| C | A | 4 | 2 | 400 | 700 | 6 | 10 | 6 | 11 | 1 |
| D | B | 6 | 4 | 600 | 1000 | 5 | 11 | 5 | 11 | – |
| E | C,D | 3 | 2 | 300 | 500 | 11 | 14 | 11 | 14 | – |

Network :- T.C. = 2400




C.P. :-

B → D → E

finding cost slope of Each activity.

C.S. for A = $\frac{1000-600}{6-4}$ = 200/day.

B = $\frac{900-500}{5-3}$ = 200/day.

C = $\frac{700-400}{4-2}$ = 150/day.

D = $\frac{1000-600}{6-4}$ = 200/day.

E = $\frac{500-300}{3-2}$ = 200/day.

Total cost = 2400,

*** Crashing is always applied on critical path as it reduce the project time and increases cost of project.

B ⟶ D ⟶ E. (C.P.)

Step ⓪ Normal schedule.

Duration ⇒ 14 days (B → D → E)

Total cost ⇒ ~~$~~ 2400

Step ① Crashing to 13 Days.

B = 200/day
D = 200/day
E = 200/day
] → All same.

→ choose minimum crash-cost slope from C.P. Activity for crashing.

Took B,

New cost (13 days) = 2400 + 200
= 2600.

↓

crashed B means Boost activity, 5 days → 4 days complete.

Step ② Crashing to 12 Days.

$\cancel{B} \rightarrow D \rightarrow E$

Crashed

↳ choose E, because E is common in both path $\left(\begin{array}{l} A \rightarrow C \rightarrow E \\ B \rightarrow D \rightarrow E \end{array}\right)$

crashed E (12days)

⇒ 2600 + 200

⇒ 2800.

→ Crashing a ~~single~~ ~~Activity~~ / common activity is more beneficial than crashing a individual acitivity.

Therefore,

Min. Duration = 12 days.

Mini cost = 2800

Ans.